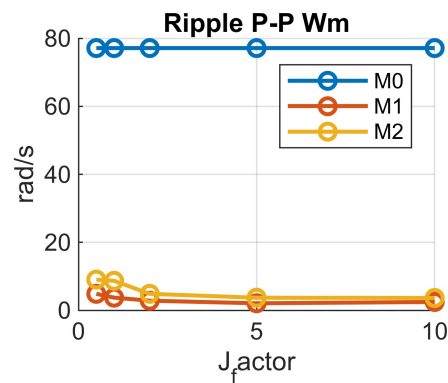
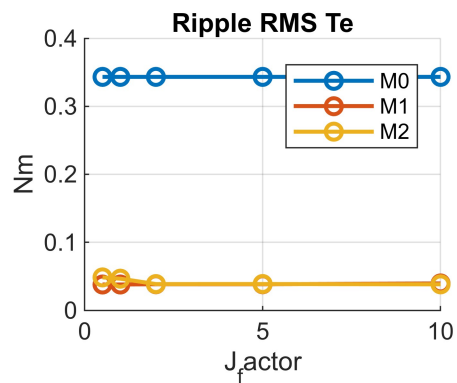
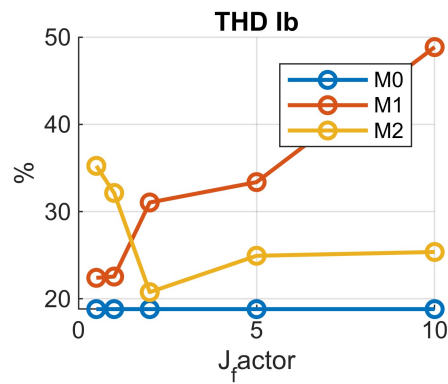
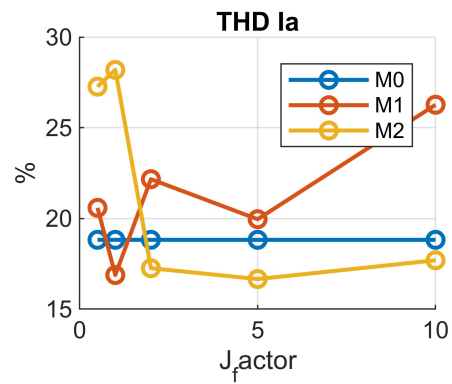


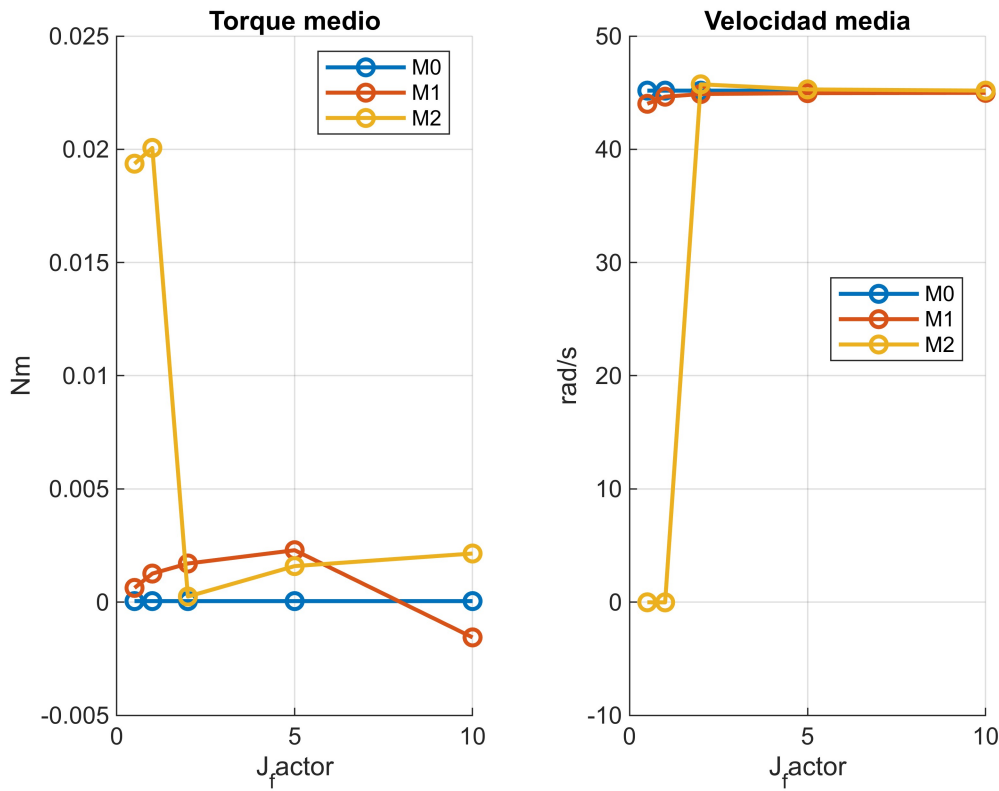
```
clc
close all
clear
```

```
graficarComparacionMotores("barrido_J", "J_factor");
```

Mostrando tabla de: barrido_J

Modelo	Motor	J_factor	J_var	fe_Hz	THD_Ia_pct	THD_Ib_pct	Te_mean	Te_ripple_rms
"Jx0.5"	"M0"	0.5	2.35e-06	359.55	18.822	18.822	4.5201e-05	0.34329
"Jx0.5"	"M1"	0.5	2.35e-06	350.31	20.584	22.408	0.00062733	0.037965
"Jx0.5"	"M2"	0.5	2.35e-06	-0.13705	27.269	35.24	0.019365	0.04842
"Jx1"	"M0"	1	4.7e-06	359.55	18.822	18.822	4.5201e-05	0.34329
"Jx1"	"M1"	1	4.7e-06	355.3	16.882	22.552	0.0012606	0.037977
"Jx1"	"M2"	1	4.7e-06	-0.17871	28.193	32.137	0.020061	0.046527
"Jx2"	"M0"	2	9.4e-06	359.55	18.822	18.822	4.5201e-05	0.34329
"Jx2"	"M1"	2	9.4e-06	357.17	22.166	31.044	0.001705	0.038364
"Jx2"	"M2"	2	9.4e-06	363.99	17.257	20.765	0.00024627	0.038635
"Jx5"	"M0"	5	2.35e-05	359.55	18.822	18.822	4.5201e-05	0.34329
"Jx5"	"M1"	5	2.35e-05	357.79	19.958	33.379	0.002292	0.038344
"Jx5"	"M2"	5	2.35e-05	360.42	16.658	24.935	0.0015907	0.038706
"Jx10"	"M0"	10	4.7e-05	359.55	18.822	18.822	4.5201e-05	0.34329
"Jx10"	"M1"	10	4.7e-05	358	26.271	48.856	-0.0015585	0.039553
"Jx10"	"M2"	10	4.7e-05	359.56	17.694	25.37	0.0021466	0.038192

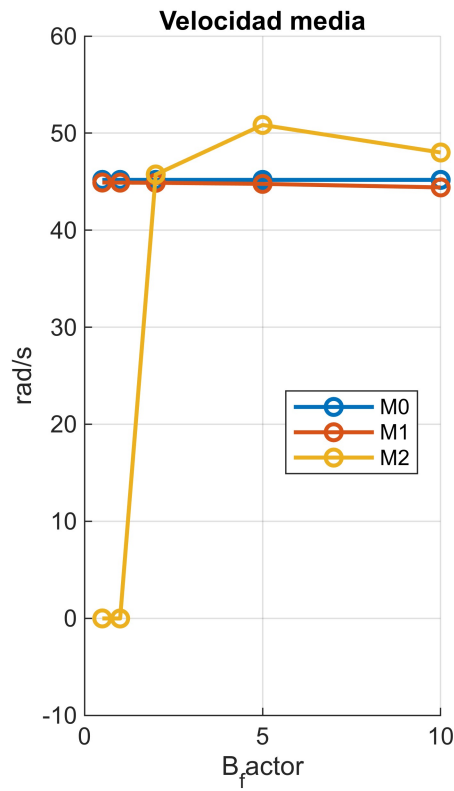
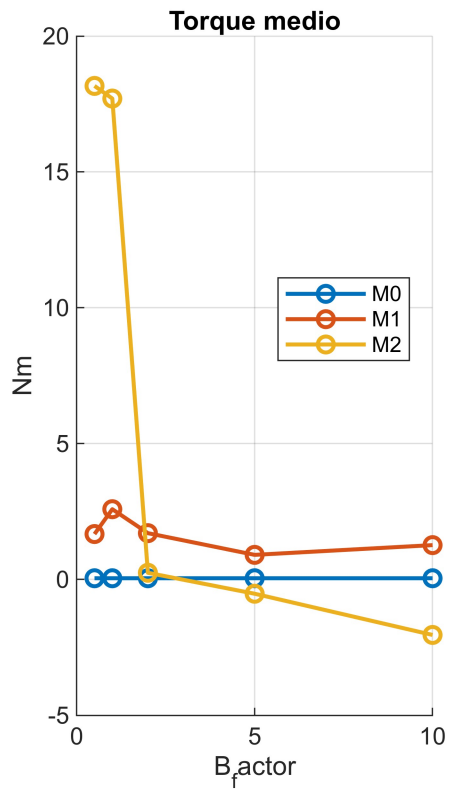
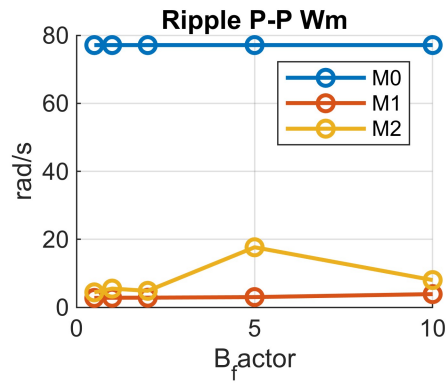
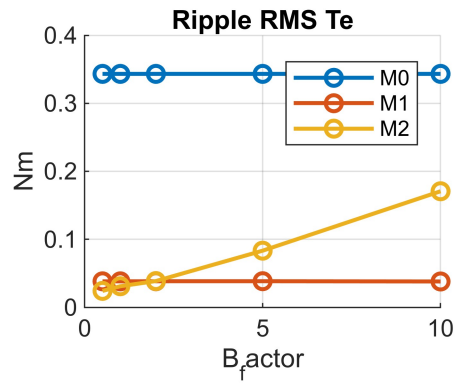
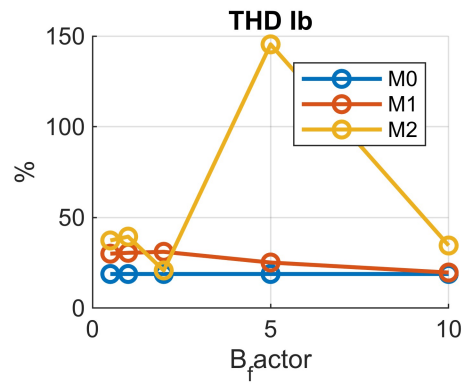
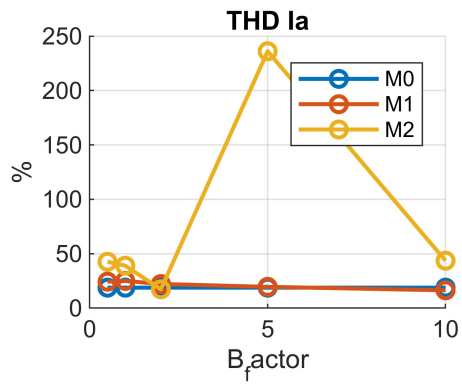




```
graficarComparacionMotores("barrido_B", "B_factor");
```

Mostrando tabla de: barrido_B

Modelo	Motor	B_factor	B_var	fe_Hz	THD_Ia_pct	THD_Ib_pct	Te_mean	Te_ripple_rms
"Bx0.5"	"M0"	0.5	0.00025	359.55	18.822	18.822	4.5201e-05	0.34329
"Bx0.5"	"M1"	0.5	0.00025	357.38	24.305	30.024	0.0016726	0.038221
"Bx0.5"	"M2"	0.5	0.00025	-0.11295	42.615	37.303	0.018167	0.02415
"Bx1"	"M0"	1	0.0005	359.55	18.822	18.822	4.5201e-05	0.34329
"Bx1"	"M1"	1	0.0005	357.32	24.971	30.494	0.0025867	0.03821
"Bx1"	"M2"	1	0.0005	-0.1507	38.976	39.426	0.017701	0.030998
"Bx2"	"M0"	2	0.001	359.55	18.822	18.822	4.5201e-05	0.34329
"Bx2"	"M1"	2	0.001	357.17	22.166	31.044	0.001705	0.038364
"Bx2"	"M2"	2	0.001	363.99	17.257	20.765	0.00024627	0.038635
"Bx5"	"M0"	5	0.0025	359.55	18.822	18.822	4.5201e-05	0.34329
"Bx5"	"M1"	5	0.0025	356.28	19.544	25.142	0.00090608	0.038408
"Bx5"	"M2"	5	0.0025	404.6	236.07	145.45	-0.00052437	0.083307
"Bx10"	"M0"	10	0.005	359.55	18.822	18.822	4.5201e-05	0.34329
"Bx10"	"M1"	10	0.005	353.39	16.357	19.619	0.0012645	0.038011
"Bx10"	"M2"	10	0.005	382.02	43.565	34.528	-0.0020416	0.17069



```
function graficarComparacionMotores(rootFolder, sortVar)
```

```

summaryFile = fullfile(rootFolder, "summaryTable.mat");
S = load(summaryFile);
summaryTable = S.summaryTable;

disp("Mostrando tabla de: " + rootFolder)
disp(summaryTable)

motorNames = unique(summaryTable.Motor);

% Figura resumen
figure('Name', "Comparación motores - " + rootFolder)
tiledlayout(2,2)

nexttile
hold on
for m = 1:numel(motorNames)
    mot = motorNames(m);
    Tm = summaryTable(summaryTable.Motor == mot, :);
    Tm = sortrows(Tm, sortVar);
    x = Tm.(sortVar);
    plot(x, Tm.THD_Ia_pct, '-o', 'LineWidth', 1.5, 'DisplayName', string(mot))
end
title("THD Ia")
xlabel(sortVar)
ylabel('%')
grid on
legend('Location','best')
hold off

nexttile
hold on
for m = 1:numel(motorNames)
    mot = motorNames(m);
    Tm = summaryTable(summaryTable.Motor == mot, :);
    Tm = sortrows(Tm, sortVar);
    x = Tm.(sortVar);
    plot(x, Tm.THD_Ib_pct, '-o', 'LineWidth', 1.5, 'DisplayName', string(mot))
end
title("THD Ib")
xlabel(sortVar)
ylabel('%')
grid on
legend('Location','best')
hold off

nexttile
hold on
for m = 1:numel(motorNames)
    mot = motorNames(m);
    Tm = summaryTable(summaryTable.Motor == mot, :);

```

```

        Tm = sortrows(Tm, sortVar);
        x = Tm.(sortVar);
        plot(x, Tm.Te_ripple_rms, '-o', 'LineWidth', 1.5, 'DisplayName',
string(mot))
    end
    title("Ripple RMS Te")
    xlabel(sortVar)
    ylabel('Nm')
    grid on
    legend('Location','best')
    hold off

    nexttile
    hold on
    for m = 1:numel(motorNames)
        mot = motorNames(m);
        Tm = summaryTable(summaryTable.Motor == mot, :);
        Tm = sortrows(Tm, sortVar);
        x = Tm.(sortVar);
        plot(x, Tm.Wm_ripple_pp, '-o', 'LineWidth', 1.5, 'DisplayName', string(mot))
    end
    title("Ripple P-P Wm")
    xlabel(sortVar)
    ylabel('rad/s')
    grid on
    legend('Location','best')
    hold off

    % Figura medias
    figure('Name', "Comparación medias - " + rootFolder)
    tiledlayout(1,2)

    nexttile
    hold on
    for m = 1:numel(motorNames)
        mot = motorNames(m);
        Tm = summaryTable(summaryTable.Motor == mot, :);
        Tm = sortrows(Tm, sortVar);
        x = Tm.(sortVar);
        plot(x, Tm.Te_mean, '-o', 'LineWidth', 1.5, 'DisplayName', string(mot))
    end
    title("Torque medio")
    xlabel(sortVar)
    ylabel('Nm')
    grid on
    legend('Location','best')
    hold off

    nexttile
    hold on

```

```

for m = 1:numel(motorNames)
    mot = motorNames(m);
    Tm = summaryTable(summaryTable.Motor == mot, :);
    Tm = sortrows(Tm, sortVar);
    x = Tm.(sortVar);
    plot(x, Tm.Wm_mean, '-o', 'LineWidth', 1.5, 'DisplayName', string(mot))
end
title("Velocidad media")
xlabel(sortVar)
ylabel('rad/s')
grid on
legend('Location','best')
hold off

```

```

end

```